LOW-RANK APPROXIMATION

The Eckart–Young–Mirsky Theorem

Truncating the SVD gives the closest matrix of lower rank.

THEOREM

Among all matrices of rank at most k, A_k is the closest to A, in the operator norm and in the Frobenius norm.

1 **TRUNCATE**

$A_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$: keep the first k singular value triples.

2 **ERROR, OPERATOR NORM**

$\|A - A_k\|_2 = \sigma_{k+1}$, the first discarded singular value.

3 **ERROR, FROBENIUS NORM**

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}.$$

4 **DECIDE k**

When $\sigma_1 \gg \sigma_2 \gg \cdots$, a few terms capture most of A: image compression, noise reduction, latent semantic analysis.